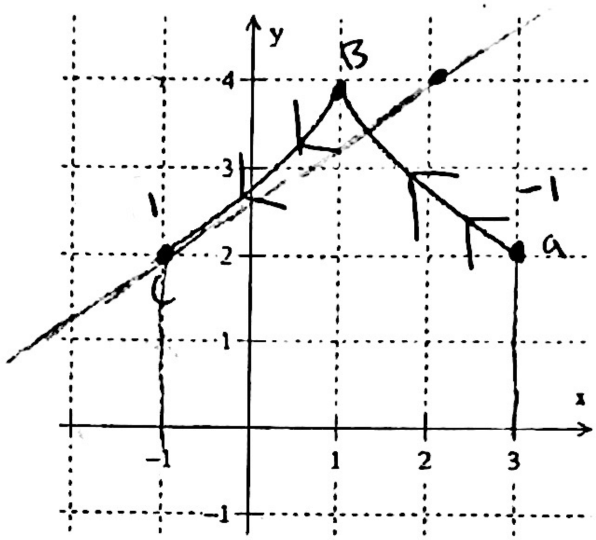
1. [12 points] Given equation and graph of the parametric curve $x = 1 - 2t^3,\ y = 4 - 2t^2;\ -1 \le t \le 1$

a. [1 pt] Find $\frac{dy}{dx}$ in terms of t

$\boxed{\frac{dy}{dx} = \frac{2}{3t}}$ work on Paper



b. [2 pts] Find $\frac{d^2y}{dx^2}$ in terms of t

$\boxed{\frac{d^2y}{dx^2} = \frac{1}{9t^4}}$ work on Paper

c. [1 pt] Draw the arrow to indicate direction of the graph.

d. [4 pts] Find equation of the tangent line to the curve at the point $t = 1$. Graph the tangent line.

$\boxed{y = \frac{2}{3}x + \frac{8}{3}}$ work on Paper

e. [1 pt] Set up the integral that represents the length of the curve. Do not evaluate the integral.

$$\int_{-1}^{1} \sqrt{(-6t^2)^2 + (-4t)^2}\, dt$$

f. [3 pts] Find the area of the region enclosed by the curve and the $x - axis$.

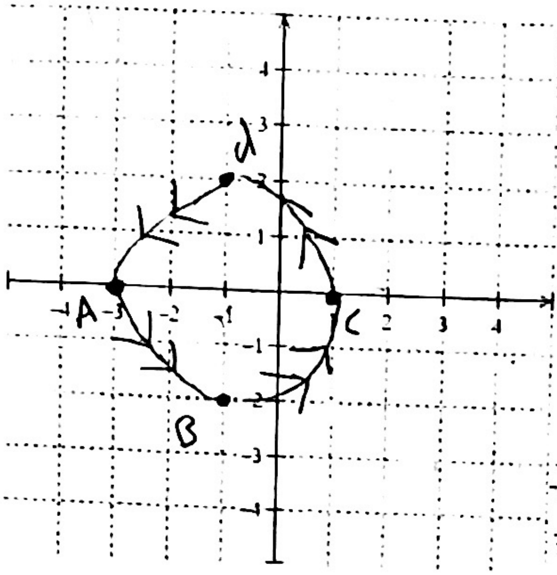
$\boxed{\frac{56}{5}}$ Work on Paper

V. good work!

+12

2. [4 pts] Find parametric equations of a particle that moves <u>once counter-clockwise</u> along the circle $(x+1)^2 + y^2 = 4$, starting at $(-3, 0)$. Include the interval in your solution and sketch the graph with arrows.

+4



$(x+1)^2 + y^2 = 4 \Rightarrow \frac{(x+1)^2}{4} + \frac{y^2}{4} = 1$

$\frac{(x+1)^2}{4} = \cos^2\theta \Rightarrow \sqrt{(x+1)^2} = \sqrt{4\cos^2\theta}$

$x = 2\cos\theta - 1$

$\frac{y^2}{4} = \sin^2\theta \Rightarrow \sqrt{y^2} = \sqrt{4\sin^2\theta}$

$y = 2\sin\theta$

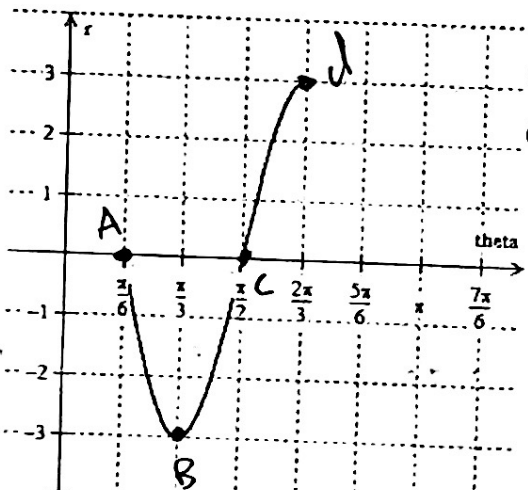
$\boxed{\pi \le t \le 3\pi}$

t	x	y	
0	1	0	
$\pi/2$	-1	2	
π	-3	0	(-3,0) a
$3\pi/2$	-1	-2	(-1,-2) B
2π	1	0	(1,0) C
$5\pi/2$	-1	2	(-1,2) d
3π	-3	0	(-3,0) e

V. good work!

3. [3 points] Sketch the curve with the given polar equation by using the graph of r as a function of θ in cartesian coordinates for $\frac{\pi}{6} \le \theta \le \frac{2\pi}{3}$. Remember to include arrow in your sketch.

+3

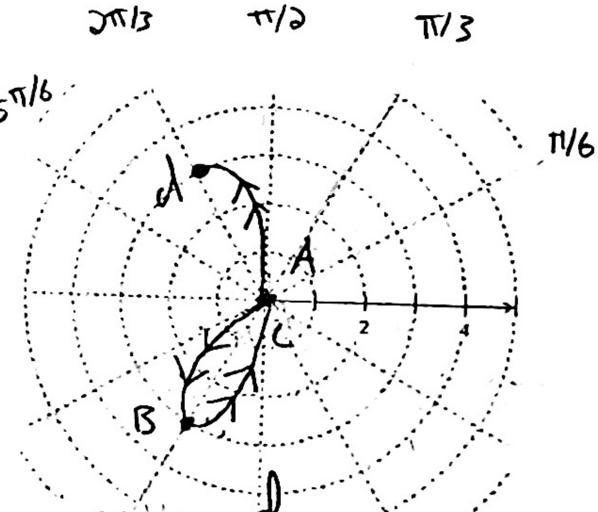


$(0, \frac{\pi}{6})$ A

$(-3, \frac{\pi}{3})$ B

$(0, \frac{\pi}{2})$ C

$(3, \frac{2\pi}{3})$ d



V. good

4. [3 points] Convert the polar equation $r\sin 2\theta = 2\sin\theta$ into cartesian equation

$x = r\cos\theta$

$y = r\sin\theta$

$\sin 2\theta = 2\sin\theta\cos\theta$

$r\sin 2\theta = 2\sin\theta$

$r(2\sin\theta\cos\theta) = 2\left(\frac{y}{r}\right)$

+3

$r(2\sin\theta\cos\theta) = \frac{2y}{r} = r^2(2\sin\theta\cos\theta) = 2y$

$2r\sin\theta \cdot r\cos\theta = 2y$

$\cancel{2}\cancel{y}x = \cancel{2y}$

$\boxed{x = 1}$

V. good.

5. [5 points] Find the slope of the tangent line to the curve $r = 4\sin\theta$ at $\theta = \pi/3$

$x = r\cos\theta \Rightarrow (4\sin\theta)\cdot(\cos\theta) \Rightarrow 4\sin\theta\cos\theta$

$y = r\sin\theta \Rightarrow (4\sin\theta)\cdot\sin\theta \Rightarrow 4\sin^2\theta$

$\frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} =$

$x = 4\sin\theta\cos\theta$

$F = 4\sin\theta \quad g = \cos\theta$

$\frac{df}{d\theta} = 4\cos\theta \quad \frac{dg}{d\theta} = -\sin\theta$

$\frac{dx}{d\theta} = (4\cos\theta)\cdot(\cos\theta) + (4\sin\theta)(-\sin\theta)$

$\frac{dx}{d\theta} = 4\cos^2\theta - 4\sin^2\theta$

$\frac{dy}{d\theta} = 8\sin\theta\cdot\cos\theta$

+5

$\frac{dy}{dx} = \frac{8\sin\theta\cos\theta}{4\cos^2\theta - 4\sin^2\theta} \Rightarrow \frac{8\sin(\pi/3)\cos(\pi/3)}{4\cos^2(\pi/3) - 4\sin^2(\pi/3)} = \frac{2\sqrt{3}}{-2} = \boxed{-\sqrt{3}}$

V. good

6. [5 points] Given $r = 24\cos(3\theta)$ set up the area bounded by the curve for $\frac{\pi}{2} \le \theta \le \pi$ as an integral and evaluate the integral.

$\int_{\pi/2}^{\pi} \frac{1}{2} r^2 \Rightarrow \frac{1}{2}\int_{\pi/2}^{\pi} (24\cos(3\theta))^2 d\theta$

+4

$\frac{576}{2}\int_{\pi/2}^{\pi} \cos^2(3\theta)\, d\theta \Rightarrow \frac{576}{2}\int_{\pi/2}^{\pi} \cos^2(u)\frac{du}{3}$

$u = 3\theta$

$du = 3\,d\theta$

$\frac{du}{3} = d\theta$

$\frac{576}{6}\int_{\pi/2}^{\pi} \frac{1}{2} + \frac{1}{2}\cos(2\theta) \Rightarrow \frac{576}{6}\int_{\pi/2}^{\pi} \frac{1}{2} + \frac{1}{2}\cos(u)\frac{du}{2}$

$u = 2\theta$

$du = 2\,d\theta$

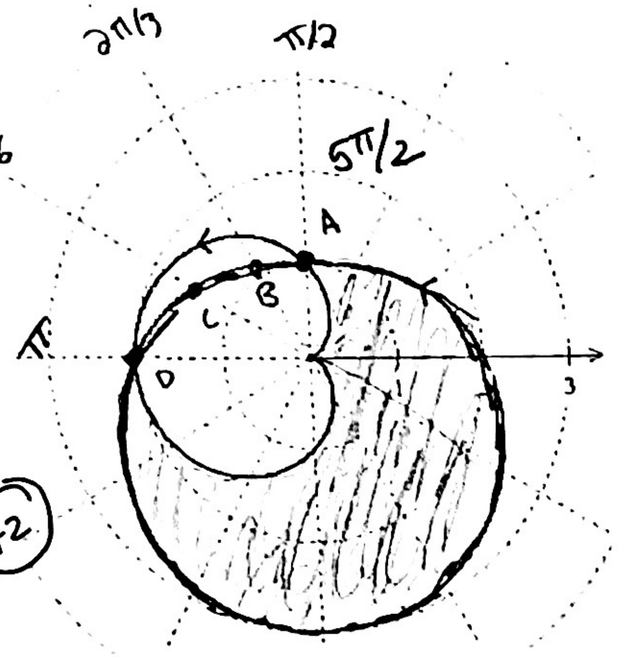
$\frac{du}{2} = d\theta$

$\frac{576}{12}\int_{\pi/2}^{\pi} \frac{1}{2} + \frac{1}{2}\cos(u) \Rightarrow \frac{576}{12}\left[\frac{1}{2}\theta + \frac{1}{2}\frac{\sin(6\theta)}{2}\right]_{\pi/2}^{\pi}$

$\frac{576}{12}\left[\frac{\pi}{2} + \frac{1}{2}\sin(6\cdot\pi)\right] - \left[\frac{1}{2}\cdot\left(\frac{\pi}{2}\right) + \frac{1}{2}\sin\left(6\cdot\frac{\pi}{2}\right)\right]$

$\frac{576}{12}\left(\frac{\pi}{2} - \frac{\pi}{4}\right) = \boxed{12\pi}$

7. [4 points] Sketch the area of the region inside the curve $r = 2 - \sin(\theta)$ and outside the curve $r = 1 - \cos(\theta)$ and set up the integral that represents the area. Do not evaluate the integral.



θ	$2-\sin\theta$	
$\pi/2$	1	a
$2\pi/3$	1.13	B
$5\pi/6$	1.5	C
π	2	D

$$\int_{\pi/2}^{\pi} \frac{1}{2}(2-\sin\theta)^2\,d\theta - \int_{-\pi}^{\pi/2} \frac{1}{2}(1-\cos\theta)^2\,d\theta$$

wrong Limit

+2

8. [8 points] Evaluate the following limits using appropriate techniques.

a. $\lim\limits_{x\to 0} \frac{x - \sin x}{x \sin x}$

$\lim\limits_{x\to 0} \frac{0 - \sin(0)}{0 \cdot \sin(0)} = \frac{0}{0}$

$\frac{df}{dx}\cdot g + f\cdot\frac{dg}{df}$

LH#1

$\lim\limits_{x\to 0} \frac{1-\cos x}{\sin x + x\cos\theta} = \frac{0}{0}$

$f = x \qquad g = \sin x$

$\frac{df}{dx} = 1 \qquad \frac{dg}{dx} = \cos\theta$

$(1)(\sin x) + (x)(\cos\theta)$

LH#2

+3

$\lim\limits_{x\to 0} \frac{0 + \sin(x)}{2\cos x - x\sin x} = \boxed{\frac{0}{2}}$

$f = x \qquad g = \cos\theta$

$\frac{df}{dx} = 1 \qquad \frac{dg}{dx} = -\sin(x)$

$(1)(\cos\theta) + (x)(-\sin x)$

b. $\lim\limits_{x\to\infty} [\ln(8x-5) - \ln(2x+3)]$

+3

$LN\left(\lim\limits_{x\to\infty} \frac{8x-5}{2x+3}\right) = \frac{\infty}{\infty}$ $LN(\infty) - LN(\infty)$ $\infty - \infty$

$LN\left(\lim\limits_{x\to\infty} \frac{8}{2}\right) = \boxed{LN(4)}$

a) Find $\frac{dy}{dx}$

$x = 1 - 2t^3$ $\quad y = 4 - 2t^2$

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{-4t}{-6t^2} = \boxed{\frac{2}{3t}}$

$\frac{dx}{dt} = -6t^2$

$\frac{dy}{dt} = -4t$

b) Find $\frac{d^2y}{dx^2}$

$\dfrac{\text{derivative of } \frac{dy}{dx}}{\frac{dx}{dt}}$

$\frac{dy}{dx} = \frac{2}{3t} \Rightarrow \frac{d'y}{dx^2} = -\frac{2}{3t^2}$

$\frac{dx}{dt} = -6t^2$

$\frac{d^2y}{dx^2} = \frac{-\frac{2}{3t^2}}{-6t^2} = \frac{2}{3t^2} \cdot \frac{1}{6t^2}$

$= \frac{2}{18t^4} = \boxed{\frac{1}{9t^4}}$

c)

t	x	y	
-1	3	2	(3, 2) a
0	1	4	(1, 4) B
1	-1	2	(-1, 2) C

d)

$x_1 = -1$

$y_1 = 2$

$m = \frac{dy}{dx}$ @ t

$m = \frac{2}{3(1)} = \frac{2}{3}$

$y - y_1 = m(x - x_1)$

$y - 2 = \frac{2}{3}(x + 1)$

$\boxed{y = \frac{2}{3}x + \frac{8}{3}}$

f) $\int_1^{-1} y(t) \cdot \frac{dx}{dt} \cdot dt \Rightarrow \int_1^{-1} (4 - 2t^2) \cdot (-6t^2)dt \Rightarrow \int_1^{-1} -24t^2 + 12t^4 dt$

$\left[-8t^3 + \frac{12}{5}t^5\right]_1^{-1} \Rightarrow \left[-8(-1)^3 + \frac{12}{5}(-1)^5\right] - \left[-8(1)^3 + \frac{12}{5}(1)^5\right]$

$\frac{28}{5} - \left(-\frac{28}{5}\right) = \boxed{\frac{56}{5}}$

MATH-183 Exam 3a-Part-II- No Calculator

Date 4/4/2025

$\cancel{9}$. [4 pts] Given the cartesian coordinate P $(-1, \sqrt{3},)$

$\cancel{a}$. Convert to polar co-ordinates with $r > 0, 0 \le \theta \le 2\pi$ — $(2, \frac{5\pi}{3})$

b. Convert to polar co-ordinates with $r < 0, 0 \le \theta \le 2\pi$ — $(-2, \frac{2\pi}{3})$

c. Convert to polar co-ordinates with $r > 0, -2\pi \le \theta \le 0$. — $(2, -\frac{\pi}{3})$

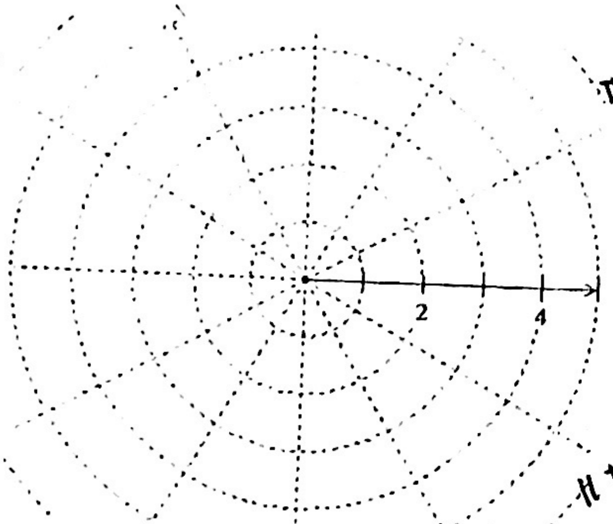
d. Convert to polar co-ordinates with $r < 0, -2\pi \le \theta \le 0$ — $(-2\ -4\pi/3)$

Wrong Quadrant

x3

$2\pi/3$ $\pi/2$ $\pi/3$

$5\pi/6$ $\pi/6$



$7\pi/6$ $11\pi/6$

$4\pi/3$ $\frac{5\pi}{3}$

$x^2+y^2=r^2 \Rightarrow \sqrt{(-1)^2+(\sqrt{3})^2} = \sqrt{1+3} = 2$

$\tan\theta = \frac{y}{x} = \frac{\sqrt{3}}{-1} = -\sqrt{3}$

$\theta = \frac{5\pi}{3}$

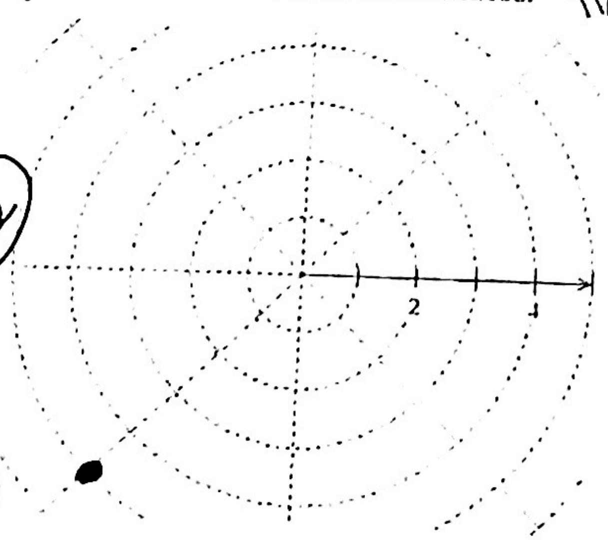
$(2, \frac{5\pi}{3})$

10. [2 points] Plot the point $P\left(-5, \frac{\pi}{4}\right)$ given in polar coordinates and convert the following polar coordinates to cartesian coordinates. $\pi/4$

$x = r\cos\theta \Rightarrow \cos(\pi/4) = \frac{\sqrt{2}}{2} \Rightarrow \frac{-5\sqrt{2}}{2}$

$y = r\sin\theta \Rightarrow \sin(\pi/4) = \frac{\sqrt{2}}{2} = -\frac{5\sqrt{2}}{2}$

x2



$(-\frac{5\sqrt{2}}{2}, -\frac{5\sqrt{2}}{2})$

4. [5 points] Find all the points of intersection of the curves. $r = 2\sin(2\theta)$ *and* $r = -1$ in the interval $0 \leq \theta \leq 2\pi$. Write the final answer as an ordered pair.

0

$2\sin(2\theta) = -1$

$\sin(2\theta) = -\frac{1}{2}$

Q III   Q IV

Talk to me about this problem!

$\theta = \frac{5\pi}{6}, \frac{7\pi}{6}$

$2\sin(2 \cdot \frac{7\pi}{6}) \Rightarrow 2\sin(\frac{7\pi}{3}) = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$

$2\sin(2 \cdot \frac{5\pi}{6}) \Rightarrow 2\sin(\frac{5\pi}{3}) = 2 \cdot \frac{\sqrt{3}}{2} = \sqrt{3}$

$(\sqrt{3}, \frac{5\pi}{6})$ and $(\sqrt{3}, \frac{7\pi}{6})$